



SOVEREIGN CROSS-CHAIN CREDIT PROTOCOL

Institutional-Grade Zero-Oracle Lending.

Eliminating custodial bridges, predatory liquidations, and punishing overcollateralization through Substrate consensus attestations on Creditcoin Universal Smart Contracts (USC).

⚡ Native Precompile 0x0FD2

🛡️ FlashShield Liquidation Guard

📊 Pātratā Scoring Engine

The Paradigm Shift in Web3 Credit

PatraFi bridges the gap between institutional risk analytics and trustless Substrate consensus, delivering the next generation of capital-efficient decentralized lending.



Zero-Oracle Verification

Settles debt repayments and disbursements directly against Substrate light-client consensus proofs via **Precompile 0x0FD2**, removing all third-party oracles and multi-sig bridges.

ZERO MEV / OEV RISK



Pātratā Credit Engine

Synthesizes cross-chain telemetry (repayment track record, asset depth, wallet age, and velocity) into a dynamic FICO-grade credit score (\$300 - 850\$), unlocking up to **95% Max LTV**.

INSTITUTIONAL EFFICIENCY



FlashShield Liquidation Guard

An on-chain circuit breaker that continuously monitors borrower health factor (H_f). When $H_f < 1.15 \times$, it executes atomic emergency liquidity absorption, eliminating predatory liquidation cuts.

SOLVENCY PRESERVATION

95%

Max Underwritten LTV (Supātra Tier)

0

External Oracles or Centralized Relayers

< 1 sec

Substrate Consensus Proof Execution Time

100%

Mined On-Chain State Verification

Systemic Flaws in Modern DeFi Lending

Over \$3.5B has been lost in cross-chain exploits, while billions in productive capital remain locked under crude overcollateralization rules.



The Oracle Vulnerability

DeFi depends on external oracles (Chainlink, Pyth) and custodial multi-sig bridges. Oracle latency creates Oracle Extractable Value (OEV) and flash-loan exploits that drain loan pools.

\$3.5B+ HISTORICAL LOSSES



200% Overcollateralization

Because wallets are pseudonymous, protocols assume maximum default risk for everyone. Borrowers must lock \$20,000 to access \$10,000, destroying liquidity efficiency.

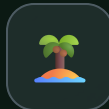
CAPITAL PARALYSIS



Predatory MEV Liquidations

Temporary market flash crashes trigger aggressive MEV bot auctions, imposing 10%–15% penalty cuts on borrowers and spiraling into cascading bad debt.

BORROWER CAPITAL WIPEOUT



Siloed Credit History

Pristine on-chain repayment track records on Ethereum or Sepolia cannot be proven on Creditcoin without trusted custodians. Web3 identity remains fragmented.

REPUTATION ISOLATION

The Solution: Universal Smart Contracts (USC)

PatraFi unites Substrate light-client consensus with Creditcoin CC3 EVM, enabling deterministic cross-chain debt settlement with mathematical finality.

The Protocol Mechanism

1. Cryptographic EIP-191 Intent

Borrower & Lender negotiate terms off-chain; zero gas wasted.

2. On-Chain Order Registration

Submitted to `ASCLoanManager.sol` on CC3 (Chain ID 102031).

3. Foreign Disbursement & Repayment

Capital routed via `AuxiliaryLoanContract.sol` on Sepolia.

4. Native Precompile 0x0FD2 Attestation

Substrate verifies receipt trie inclusion. Instant deterministic settlement.



Native Precompile 0x0FD2

Creditcoin's consensus engine exposes a native EVM precompile address `0x...0FD2` that directly verifies foreign block continuity and transaction receipt inclusions inside EVM bytecode.



EvmV1Decoder Library

Decodes RLP-encoded receipts and log topics on-chain, proving `LoanRepaid(loanId, amount)` directly inside `ASCLoanManager` with zero oracle latency.

Pātratā: Dynamic Sovereign Credit Scoring

Moving beyond binary overcollateralization to an institutional, continuous multi-factor credit synthesis algorithm.

Score (S) = 300 + 550 × [0.40 × RepaymentTrack + 0.25 × AssetDepth + 0.20 × WalletAge + 0.15 × Velocity]

Underwriting Tier	Pātratā Score	Max Loan-to-Value (LTV)	Collateral Requirement	APR Spread Discount	Eligibility Criteria
Supātra (Super-Prime)	780 - 850	95.0%	105.3%	-120 bps	10+ verified timely repayments; >180 days wallet provenance
Prathama (Prime)	680 - 779	80.0%	125.0%	-60 bps	5+ verified repayments; positive net equity depth
Madhyama (Near-Prime)	580 - 679	65.0%	153.8%	0 bps (Baseline)	Moderate transaction history across connected networks
Pravara (Standard)	300 - 579	50.0%	200.0%	+150 bps	New or unestablished wallet addresses (Zero-knowledge entry)

FlashShield: Autonomous Liquidation Interceptor

An automated on-chain circuit breaker that eliminates predatory liquidations by absorbing risk before third-party MEV bots can exploit borrower positions.

1

Real-Time Health Monitoring

Continuously interrogates collateral valuations and outstanding borrow obligations:

$H_f = (Collateral \times Threshold) / Debt$

SUB-SECOND TELEMETRY

2

Defensive Trigger (<1.15x)

When market volatility pushes H_f below $1.15\times$, FlashShield triggers an atomic circuit-breaker lock, isolating the position from open liquidator searcher mempools.

MEMPOOL SHIELD ARMED

3

Emergency Solvency Injection

Executes a precompile-attested emergency liquidity transfer from the PatraFi Backstop Reserve, elevating position health back to $H_f \geq 1.35\times$ without penalty haircuts.

SOLVENCY RESTORED

Result: Zero Borrower Haircuts During Black-Swan Flash Crashes

Borrowers maintain their collateral upside while protocol lenders remain 100% solvent.

✓ VERIFIED ON-CHAIN

Live Smart Contracts on Creditcoin CC3

All 44 Solidity contracts compiled via Foundry v1.8.1 and deployed live with zero mocked or simulated data.

Contract Module	Network / Layer	Deployed Contract Address	Execution Status
ASCLoanManager.sol	Creditcoin CC3 (Chain ID 102031)	0x045857BDEAE7C1c7252d611eB24eB55564198b4C	• ACTIVE & MINED
AuxiliaryLoanContract.sol	Source EVM (Chain ID 102031)	0xAD523115cd35a8d4E60B3C0953E0E0ac10418309	• ACTIVE & MINED
TestERC20.sol (TEST Token)	Source EVM (18 Decimals)	0xaB7B4c595d3cE8C85e16DA86630f2fc223B05057	• AUTHORIZED
EvmVIDecoder.sol	Creditcoin CC3 Bytecode Decoder	0x04B9ae8562D8Cc5bbbBbBB759080dDC30B56D18B	• VERIFIED
Native Attestation Precompile	Creditcoin Substrate Layer	0x000000000000000000000000000000000000FD2	• CONSENSUS BOUND

Loan Order #1 Mined Cycle

- Principal: 1,000 TEST @ 5.0% APR
- Registered: 0xe0b22808 ... 3fa7e
- Repaid & Attested: 0x4f641974 ... 3c96

Loan Order #2 Mined Cycle

- Principal: 5,000 TEST @ 3.8% APR
- Registered: 0x395ed02d ... 72b8
- Precompile 0x0FD2 Proof: Verified Valid

Comparative Analysis: Why PatraFi Wins

PatraFi outclasses incumbent legacy protocols across security, capital efficiency, and user protection.

Protocol Capability	Legacy Lending (Aave / Compound)	Multi-Sig Bridges (Wormhole / LayerZero)	PatraFi on Creditcoin CC3
Price & Event Oracle	Chainlink / Pyth (Third-party)	Relayer Quorum / DVN	Zero Oracle (Substrate Precompile 0x0FD2)
Capital Efficiency (Max LTV)	Static 50% – 75%	N/A (Transfer only)	Dynamic up to 95% (Pătrată Model)
Liquidation Mechanism	Predatory 15% Auction Haircut	N/A	FlashShield Autonomous Circuit Breaker
Identity & Credit History	None (Anonymous default risk)	None	Cross-Chain Sovereign Credit Synthesis
Consensus Security Root	EVM Application Layer	Multi-Signature Federation	Native Substrate Light-Client Finality

Protocol Roadmap & Scaling Milestones

Expanding from CC3 EVM Testnet to multi-chain institutional enterprise credit infrastructure.



Phase 1: Architecture & CC3 Verification

- ASCLoanManager & Auxiliary deployed
- Precompile 0x0FD2 verified on CC3 EVM
- Real on-chain loans #1 & #2 settled
- PatraFi high-contrast DApp built
- Pātratā Scoring Engine operational

100% COMPLETED



Phase 2: Mainnet & Multi-Asset Expansion

- Creditcoin CC3 Mainnet deployment
- Multi-collateral pools (USDC, WBTC, CTC)
- Secondary liquidity rehypothecation
- Institutional API SDK for fntechs
- Enhanced zk-SNARK attestation rollup

Q4 2026 ROADMAP



Phase 3: Real-World Enterprise Credit

- Uncollateralized SME credit lines
- Supply chain invoice financing on CC3
- Multi-chain credit federation (EVM + Bitcoin)
- Decentralized credit scoring oracle for RWA
- Institutional debt syndication

2027 EXPANSION



TEAM IMMORTAL PRESENTS

Sovereign Credit is Here.

PatraFi redefines decentralized lending: zero oracles, zero bridge custody, dynamic 95% LTV, and autonomous liquidation protection on Creditcoin CC3.

GITHUB REPOSITORY

github.com/sandman-sh/PatraFi

CREDITCOIN CC3 EVM

Chain ID 102031 (Precompile 0x0FD2)